

GOODWELL 2 E, OK, USA

WMO: 723580

Lat: 36.5993N

Lon: 101.5950W

Elev: 3266

StdP: 13.04

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03055

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.9	11.8	-2.6	5.4	18.1	1.4	6.6	27.8	25.8	37.2	22.9	39.0	6.6	N/A	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	100.1	65.7	97.6	66.1	95.1	65.7	71.1	87.7	69.9	86.7	69.0	85.9	11.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.8	111.5	74.8	65.6	106.9	74.2	64.4	102.4	73.7	37.0	87.6	36.0	86.0	35.1	85.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	20.0	17.8	-2.6	104.7	5.0	2.3	-6.2	106.4	-9.1	107.7	-11.9	109.0	-15.5	110.7		
			-3.6	72.6	5.5	1.7	-7.6	73.8	-10.8	74.7	-13.9	75.7	-17.9	76.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	35.5	37.1	47.6	54.8	64.5	75.7	79.2	77.0	70.6	57.3	45.9	35.2	
	DBStd	18.12	10.33	11.06	9.97	9.14	8.77	6.13	5.21	5.38	7.07	9.29	9.62	10.28	
	HDD50	1740	456	376	161	53	7	0	0	0	0	37	186	464	
	HDD65	4472	913	780	541	322	116	3	1	1	30	268	574	923	
	CDD50	4225	7	16	87	197	457	772	905	836	617	264	62	5	
	CDD65	1484	0	0	2	15	101	325	441	372	198	30	0	0	
	CDH74	19232	3	21	163	492	1613	4159	5597	4240	2297	577	70	0	
	CDH80	10017	0	3	35	161	717	2269	3242	2292	1086	204	8	0	
Wind	WSAvg	8.3	7.6	8.3	9.0	10.0	9.1	9.4	8.0	7.3	8.0	7.9	7.9	7.4	
Precipitation	PrecAvg	16.7	0.3	0.4	0.9	1.2	2.8	2.7	2.6	2.2	1.6	1.1	0.6	0.3	
	PrecMax	25.6	0.9	1.6	4.8	4.8	6.3	8.7	7.1	5.4	7.3	5.8	3.3	1.9	
	PrecMin	11.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	
	PrecStd	3.2	0.3	0.4	1.1	1.1	1.7	1.7	1.7	1.5	1.5	1.3	0.8	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	77.0	84.4	90.2	96.5	103.9	103.5	101.2	97.4	91.1	80.8	70.3	
		MCWB	47.8	48.8	53.4	55.0	57.6	62.5	66.7	66.2	64.7	58.3	52.2	47.1	
	2%	DB	66.3	69.5	79.0	84.5	92.2	99.3	100.4	98.3	94.2	85.9	75.2	63.3	
		MCWB	45.5	46.9	52.4	54.1	58.8	63.8	66.6	66.7	63.7	58.5	51.3	45.2	
	5%	DB	59.4	64.1	72.8	80.2	88.0	95.7	98.0	95.6	90.8	81.0	69.3	58.1	
		MCWB	42.8	44.6	50.0	54.1	59.0	65.3	67.0	66.9	63.9	57.5	49.4	42.8	
	10%	DB	53.4	57.1	67.5	74.5	83.7	92.5	95.2	92.3	86.8	75.3	63.3	52.7	
		MCWB	40.0	41.8	48.7	53.7	59.7	65.5	66.8	66.7	63.9	55.6	47.2	40.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.1	51.0	58.9	62.1	68.2	71.2	72.5	72.7	69.7	66.2	56.4	50.3
			MCDB	69.0	70.4	73.3	73.8	80.7	89.6	90.0	86.9	85.7	76.5	67.0	60.6
2%		WB	45.9	48.1	55.7	59.1	65.9	69.9	70.9	71.2	68.2	63.2	53.2	46.2	
		MCDB	63.4	70.0	71.2	71.8	78.6	87.4	88.5	86.4	84.0	73.8	68.4	60.7	
5%		WB	43.0	45.5	52.6	57.1	64.1	68.8	69.7	69.9	67.0	60.5	50.7	43.2	
		MCDB	57.2	64.1	68.9	70.7	77.6	86.0	87.0	86.0	82.7	74.2	67.3	54.8	
10%		WB	40.3	41.7	49.6	54.9	61.9	67.8	68.9	68.8	65.8	57.7	47.9	40.8	
		MCDB	52.2	56.0	67.6	69.4	76.2	84.9	86.5	85.1	81.1	71.4	62.8	51.7	
Mean Daily Temperature Range	5% DB	MDBR	27.7	28.0	29.8	28.8	28.3	28.3	27.3	26.0	26.9	28.0	28.8	25.7	
		MCDBR	37.0	39.8	37.9	38.4	35.0	33.2	31.7	31.1	32.2	35.7	37.0	33.8	
		MCWBR	20.4	21.0	17.9	16.7	13.3	9.9	7.9	7.9	9.8	14.4	17.5	19.1	
	5% WB	MCDBR	34.6	38.7	34.6	28.8	28.7	27.8	27.4	26.8	25.8	28.7	32.1	31.1	
		MCWBR	19.5	20.6	17.6	14.7	12.4	9.5	7.9	8.5	8.8	13.2	16.5	18.5	
Clear-Sky Solar Irradiance	taub		0.255	0.263	0.291	0.320	0.338	0.372	0.376	0.362	0.336	0.299	0.270	0.255	
	taud		2.581	2.571	2.497	2.432	2.441	2.386	2.416	2.424	2.480	2.555	2.563	2.585	
	Ebn at Noon		305	314	312	305	299	287	285	288	292	296	296	298	
	Edh at Noon		23	27	31	36	36	38	37	36	32	27	23	22	
All-Sky Solar Radiation	RadAvg		915	1177	1578	1965	2242	2400	2338	2059	1754	1315	999	804	
	RadStd		81	112	130	108	134	117	101	104	82	141	89	56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page